Chapter 1 Review Solutions

Computer Networking: A Top-Down Approach (9th Ed.)

Chapter 1 Review Questions — Full Solutions

All 28 questions (R1–R28) are solved below — concept explanations plus, wherever a calculation is involved, a clear step-by-step derivation.

Section 1.1 — Networks, Internet, Protocols

R1. Host vs End System

Key point: A host and an end system are the **same thing** — the textbook uses these terms interchangeably. Both refer to any device that sits at the **edge** of the network and runs applications.

Types of end systems: desktop/laptop computers, smartphones, servers, IoT devices (smart cameras, thermostats), gaming consoles, smart TVs.

Is a Web server an end system? Yes — it is a host that sits at the network edge and runs the HTTP server application. This is different from network "core" devices (routers/switches), which only forward packets and do not run end-user applications.

R2. ChatGPT's description of diplomatic protocol

Diplomatic protocol refers to the established rules, customs, and formal procedures that nations and diplomats follow when interacting with one another — such as the correct order of introductions, the format of correspondence, rules of precedence, and treaty-signing procedures. The core idea is that **both parties know in advance exactly how the interaction will unfold**, which avoids confusion.

This same idea maps directly onto a network protocol: a network protocol defines the format of messages, the order in which they are exchanged, and the actions to be taken on receipt — so that the sender and receiver can understand each other without ambiguity.

R3. Why standards are important

Standards are essential because different vendors build different hardware and software. If every vendor designed its own protocol, Host A (built by vendor X) would not be able to communicate with Host B (built by vendor Y).

Standards (such as IETF RFCs, IEEE, and ITU specifications) guarantee **interoperability** — regardless of which vendor's equipment is used, communication is guaranteed to work correctly.

Section 1.2 — Network Edge & Access Technologies

R4. Four access technologies (classified)

Technology	Classification
DSL	Home access
Ethernet LAN	Enterprise access
Wi-Fi	Home / Enterprise access
4G / 5G Cellular	Wide-area wireless access

R5. Is HFC transmission rate dedicated or shared?

HFC (Hybrid Fiber Coax) transmission rate is **shared** — all users in a neighborhood share the same coaxial cable segment.

Are downstream collisions possible? No — downstream traffic flows in only one direction: from the cable head-end to homes (broadcast-style). Only the head-end transmits on that channel, so there is only a single sender and therefore no possibility of collision.

(Collisions *can* occur upstream, since multiple homes may try to transmit toward the head-end at the same time.)

R6. Residential access technologies in your city

This is a personal/local research question — it depends on your specific city and the ISPs available there, and pricing changes over time, so there is no single fixed answer. To complete this, check the websites of your local ISPs and note, for each access type, the advertised downstream rate, upstream rate, and monthly price. This question is designed for you to research yourself.

R7. Ethernet LAN transmission rates

Ethernet has evolved through many speed generations: 10 Mbps → 100 Mbps (Fast Ethernet) → 1 Gbps (Gigabit Ethernet) → 10 Gbps → 40/100 Gbps, with current standards reaching up to 400 Gbps.

R8. Physical media that Ethernet can run over

- Twisted-pair copper cable (UTP — Cat5e, Cat6)
- Coaxial cable (older installations)
- Fiber-optic cable (high-speed, long-distance)

R9. HFC, DSL, FTTH, FWA, LEO satellite — rates and shared/dedicated

Technology	Typical Rate Range	Shared / Dedicated
DSL	A few Mbps – ~100 Mbps	Dedicated line (per subscriber)
HFC (Cable)	10s of Mbps – 1+ Gbps downstream	Shared among neighborhood
FTTH	100s of Mbps – multi-Gbps	Mostly dedicated (PON split among a few users)
FWA (Fixed Wireless Access)	10s – 100s of Mbps	Shared spectrum
LEO Satellite	~50 – 200+ Mbps	Shared among users in the coverage cell

R10. Popular wireless Internet access technologies today

- **Wi-Fi**: short range, unlicensed spectrum, high bandwidth, inexpensive — well suited to home/office hotspots.
- **Cellular (4G/5G)**: wide-area coverage, supports mobility, licensed spectrum, cost tied to data usage.
- **LEO Satellite** (e.g., Starlink-type systems): covers remote/rural areas that wired or cellular infrastructure cannot reach; latency is higher than terrestrial links but far lower than older GEO satellite systems.

Comparison: Wi-Fi offers the best speed for the cost but only over short range; cellular provides mobility and wide coverage but at a higher cost per GB; satellite offers unmatched coverage in remote areas but at higher cost and latency.

Section 1.3 — Network Core

R11. Store-and-forward delay (one switch)

Key concept: A store-and-forward switch receives the **entire packet** before it forwards any of it (it does not forward bit-by-bit as data arrives).

```
Total delay = transmission delay on link 1 + transmission delay on link 2
            = L/R1 + L/R2
```

(Queuing, propagation, and processing delay are ignored, as stated in the question.)

R12. Circuit switching vs. TDM vs. FDM

Advantage of circuit switching over packet switching: For the duration of the connection, the user is guaranteed **dedicated bandwidth** — this makes performance steady and predictable (no congestion once the connection is established), which is well suited to real-time traffic.

TDM vs. FDM (both are circuit-switching techniques):

- TDM gives each user the full bandwidth of the link, but only during its assigned **time slot** — this is more efficient for digital data and requires no guard bands.
- FDM divides the frequency spectrum into bands, giving each user its own dedicated band — but **guard bands** are needed between bands to prevent interference, which wastes some of the spectrum. As a result, TDM is generally more efficient for digital systems.

R13. Statistical Multiplexing (2 Mbps link, users transmit at 1 Mbps, active 20% of the time)

a) With circuit switching, how many users can be supported?

Circuit switching must permanently reserve 1 Mbps for each user regardless of whether they are actively transmitting.

```
Users = 2 Mbps / 1 Mbps = 2 users
```

b) With packet switching, when does queuing occur?

If two or fewer users transmit at the same time, the combined demand (≤ 2 Mbps) does not exceed the link capacity (2 Mbps), so there is essentially **no queuing delay**.

If three users transmit at the same time, the combined demand is 3 Mbps, which exceeds the 2 Mbps link capacity — the excess data backs up in a queue, so **queuing delay occurs**.

c) Probability that a given user is transmitting:

```
p = 0.20
```

d) With three users, probability that all three transmit simultaneously:

```
P(all 3 active) = 0.2 × 0.2 × 0.2 = 0.008 = 0.8%
```

The queue only grows when all three users are transmitting simultaneously (since only then does demand, 3 Mbps, exceed capacity, 2 Mbps), so:

```
Fraction of time the queue grows = 0.008 = 0.8%
```

R14. Peering and IXP economics

Why do same-tier ISPs often peer with each other? Two ISPs at the same level of the hierarchy connect directly to each other (“peer”) so that traffic between their respective customers can be exchanged without paying a higher-tier (transit) provider. Because traffic tends to flow roughly equally in both directions, the two ISPs often agree to settlement-free peering — saving costs for both.

How does an IXP earn money? An Internet Exchange Point (IXP) provides shared physical infrastructure where multiple ISPs can co-locate and connect to each other. In exchange, the IXP charges **membership and port fees** — this works out cheaper for ISPs than building individual peering links to every other ISP.

R15. Google’s own network

Google has built its own **private global network** connecting its data centers, and it also peers directly with access ISPs (often at IXPs), reducing its reliance on tier-1 backbone providers.

Motivation for content providers to build their own networks: greater control over performance (lower latency, higher throughput), reduced transit costs, improved reliability, and the ability to place content closer to end users (e.g., through CDN/edge servers) for faster delivery of services like search results and video.

Section 1.4 — Delay, Loss, and Throughput

R16. Components of end-to-end delay

The four components are: **processing delay, queuing delay, transmission delay, and propagation delay**.

- **Variable:** Queuing delay — it depends on current network traffic/congestion and changes over time.
- **Roughly constant** (for a fixed route and packet size): Processing delay, transmission delay, and propagation delay — these depend on router hardware, packet length, and physical distance, which are generally fixed.

R17. Transmission delay vs. propagation delay (interactive animation)

This question references an interactive animation on the textbook's companion website, which I cannot access directly, but the underlying concept is as follows:

Key comparison: Transmission delay (L/R) vs. propagation delay (d/s).

- **The sender finishes transmitting before the first bit reaches the receiver:** This happens when the transmission rate is high and the packet is small (low transmission delay), but the distance is very large (high propagation delay) — for example, over satellite links.
- **The first bit reaches the receiver before the sender finishes transmitting:** This happens when the distance is short (low propagation delay) or the transmission rate is slow / the packet is large (high transmission delay).

You can try different rate, distance, and packet-size combinations in the animation itself to verify both scenarios.

R18. Propagation delay calculation

Given: L = 1,000 bytes, d = 2,500 km = 2,500,000 m, s = 2.5 × 10^8 m/s, R = 2 Mbps

```
Propagation delay = d / s
                  = 2,500,000 m / (2.5 × 10^8 m/s)
                  = 0.01 s
                  = 10 ms
```

Key point: The propagation delay formula depends only on **distance and propagation speed**:

```
Propagation delay = d / s
```

Does this delay depend on packet length (L)? No. **Does this delay depend on transmission rate (R)?** No.

(L and R only affect **transmission delay** = L/R, which is a separate delay component. The values of L and R given in the question are essentially "distractors," testing whether the correct propagation-delay formula is understood.)

R19. Three-link path throughput (bottleneck concept)

Given: R1 = 500 kbps, R2 = 2 Mbps, R3 = 1 Mbps

a) Throughput = the bottleneck link (minimum rate):

```
Throughput = min(500 kbps, 2000 kbps, 1000 kbps) = 500 kbps
```

b) Time to transfer a 4,000,000-byte file:

```
File size in bits = 4,000,000 × 8 = 32,000,000 bits
Time = 32,000,000 bits / 500,000 bps = 64 seconds
```

c) Repeat with R2 reduced to 100 kbps:

```
New bottleneck = min(500 kbps, 100 kbps, 1000 kbps) = 100 kbps
Time = 32,000,000 bits / 100,000 bps = 320 seconds
```

R20. Packet creation and router forwarding

Host A breaks the file into small **packets**, encapsulating each one with header information (source address, destination address, sequence number, etc.).

When a packet arrives at a router, the router examines the **destination address** in the packet header and consults its **forwarding table** to decide which outgoing link to send the packet on.

Why this is analogous to asking directions while driving: just as a driver does not know the entire route in advance and instead asks for directions at each intersection before proceeding to the next stretch of road, a router does not know the complete end-to-end path in advance either — it only decides the next hop locally, based on its own forwarding table, one packet at a time.

R21. Queuing and loss animation concept

This is also an interactive tool that I cannot access directly, but the underlying concept is:

```
Traffic Intensity = Emission (arrival) rate / Transmission (service) rate
```

If this ratio is **greater than or equal to 1**, the queue grows without bound and the buffer will eventually overflow, causing **packet loss**.

Across two separate runs, the exact time at which loss occurs may **differ** if there is any randomness or timing variability in the system — the general trend (higher traffic intensity leads to faster loss) will hold, but the exact numbers depend on the animation's internal timing. This needs to be run and observed directly in the tool.

Section 1.5 — Protocol Layers

R22. Five tasks a layer can perform

- Error detection/correction
- Flow control (matching sender and receiver speeds)
- Segmentation and reassembly
- Multiplexing/connection setup
- Addressing and routing decisions

Can more than one layer perform the same task? Yes — for example, **error checking** is performed both at the link layer (frame check sequence) and at the transport layer (TCP/UDP checksum). This "belt-and-suspenders" design adds redundancy for reliability.

R23. The five layers of the Internet protocol stack

Layer	Principal Responsibility
Application	Supports network applications (HTTP, SMTP, DNS) — defines the messages exchanged
Transport	Process-to-process data transfer (TCP = reliable, UDP = unreliable)
Network	Routes datagrams from the source host to the destination host (IP)
	Transfers frames between neighboring network elements

Link	Transfers frames between neighboring network elements over a single link (Ethernet, Wi-Fi)
Physical	Moves individual bits within a frame from one node to the next over the physical medium

R24. Data-unit terminology

- **Application-layer message**: the data unit exchanged by an application-layer protocol (e.g., an HTTP request/response).
- **Transport-layer segment**: the message plus a transport-layer header, produced by encapsulation — a TCP or UDP segment.
- **Network-layer datagram**: the segment plus a network (IP) header — an IP datagram.
- **Link-layer frame**: the datagram plus a link-layer header/trailer — the final unit transmitted over a physical link.

R25. Which layers each device processes

Device	Layers Processed
Router	Network, Link, Physical (examines up to the network layer to make forwarding decisions)
Link-layer switch	Link, Physical only
Host	All five layers (Application through Physical — the full protocol stack)

Section 1.6 — Network Security

R26. Self-replicating malware

Malicious software that can propagate itself to other hosts **without requiring any explicit user action** — such as a worm. Once it infects one host, it automatically searches for and infects other vulnerable hosts, spreading exponentially through the network.

(This differs from a virus, which typically requires some form of user interaction to spread, such as opening an infected attachment.)

R27. Creating a botnet and using it for a DDoS attack

Creating a botnet: An attacker infects a large number of hosts with malware (often via a worm or trojan), either by exploiting vulnerabilities or by tricking users into installing it. These infected hosts become "**bots**" (or "zombies") that the attacker can control remotely, typically through a command-and-control server. The collection of these compromised hosts is called a **botnet**.

Using it for DDoS: The attacker commands all of the bots to send a massive volume of traffic or requests to a target server at the same time. This overwhelms the target's resources/bandwidth, preventing legitimate users from accessing the service.

R28. Man-in-the-middle position (Trudy) — malicious actions

Since Trudy can intercept and control packets in both directions between Alice and Bob, she can:

- **Eavesdrop**: passively read/record the contents of the communication without altering anything.
- **Modify/Tamper**: alter the content of packets before forwarding them on.
- **Impersonate/Spoof**: pretend to be Bob when communicating with Alice, and pretend to be Alice when communicating with Bob, without either party knowing.
- **Inject**: insert fabricated messages that appear to come from Alice or Bob.
- **Drop/Delay**: selectively discard or delay messages (a form of denial of service).
- **Replay**: capture legitimate messages and resend them at a later time.

Note: R6, R17, and R21 are inherently interactive/personal-research questions (requiring city-specific data or the textbook's companion-website animations). For these, the underlying concept has been explained so the animation/research can be completed directly by the reader.